

Chapter 16.6: Surface Integrals

V4: Surface Integrals. I can set up and compute surface integrals for scalar and vector valued functions.

Mechanics

1. Integrate $f(x, y, z) = yz$ over the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies above the cone $z = \sqrt{x^2 + y^2}$.
2. Find the flux of the field $\mathbf{F}(x, y, z) = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$ across the surface S which is the boundary of the solid half-cylinder $0 \leq z \leq \sqrt{1 - y^2}, 0 \leq x \leq 2$, with the outward orientation.

Applications

3. A fluid has constant density 870 kg/m^3 and flows with velocity $\mathbf{v} = \langle z, y^2, x^2 \rangle$, where x, y, z are measured in meters and the components of $\mathbf{v}$ in meters per second. Find the rate of flow outward through the cylinder $x^2 + y^2 = 4, 0 \leq z \leq 1$.
4. Consider a sheet of metal that has been molded into the shape of a paraboloid $z = x^2 + y^2$, with $z \in [0, 1]$. Suppose also that its density is given by $\delta(x, y, z) = x^2 + y^2$.
 - (a) Compute the total mass of the sheet of metal.
 - (b) Suppose you wanted to cut this sheet of metal with a horizontal cut so that the two pieces you obtain have the same mass. Where should you cut? What if you wanted to make a vertical cut?
5. One of the premier trout fishing destinations in North America is Bow River, located in southern Alberta. There, marine biologists measured that the flow of rainbow trout near the northern tip of the river can be modeled by the vector field $\mathbf{F}(x, y, z) = \langle -x, -y, -z^4 \rangle$. Excited by the flourishing aquatic life, you set up a hemispherical net just under the surface of the water, given by $x^2 + y^2 + z^2 = 4$, with $z \leq 0$. Compute the approximate number of trout flowing **into** your net.

Extensions

6. Give an example of a nontrivial vector field $\mathbf{F}(x, y, z)$ and a nonempty surface S so that

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = 0$$

[Hint: This will be a much easier exercise to do after the material from next week!]